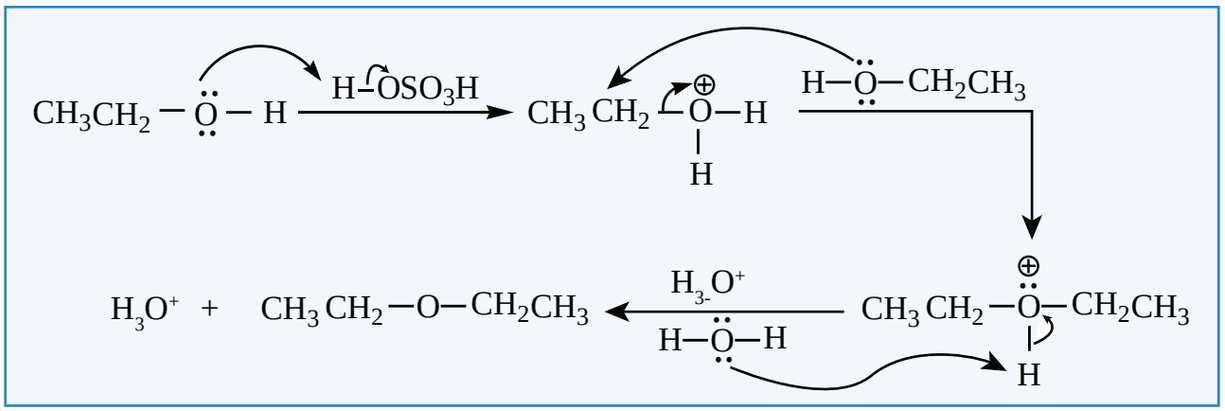


2. Williamsons synthesis:

When an alkyl halide is heated with an alcoholic solution of sodium alkoxide, the corresponding ethers are obtained. The reaction involves SN^2 mechanism.

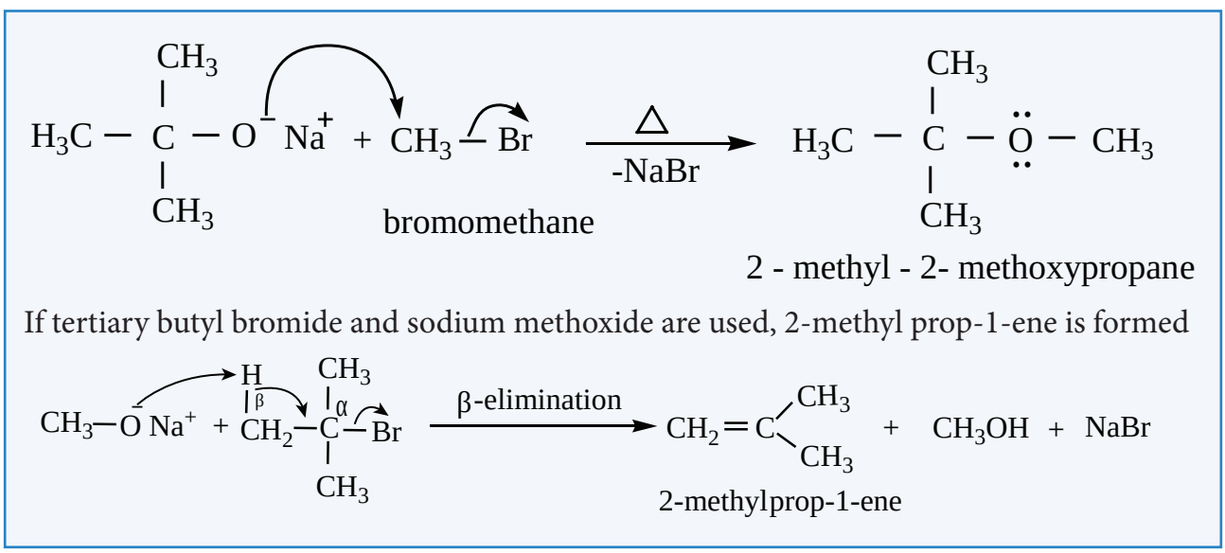
$$CH_3\text{-}ONa + Br\text{-}C_2H_5 \xrightarrow{\Delta} CH_3\text{-}O\text{-}C_2H_5 + NaBr$$

Mechanism:

$$CH_3 - O^-Na^+ + CH_3 - CH_2 - Br \xrightarrow[\Delta]{\text{-NaBr}} CH_3 - CH_2 - O - CH_3$$

methoxyethane

We know that primary alkyl halides are more susceptible for SN^2 reaction. Hence for the preparation of mixed ether having primary and tertiary alkyl group, primary alkyl halide and tertiary alkoxide are used. On the other hand, if we use tertiary alkyl halide and primary alkoxide, elimination dominates and succeeds over substitution to form an an alkene.



Methylation of alcohol

Methyl ethers can be prepared by treating an alcohol with diazomethane in presence of catalyst, fluoroboric acid.

$$CH_3\text{-}CH_2\text{-}OH + CH_2N_2 \xrightarrow[\Delta]{HBF_4} CH_3\text{-}CH_2\text{-}O\text{-}CH_3 + N_2$$

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